

Low-Complexity FO-Tolerant Frame Synchronisation Using an Amplitude-Weighted Timing Metric and QAM-Based Training Sequence for Low-SNR Coherent Optical Inter-Satellite Links

Shoma Tateno^{(1)*}, Hidemi Noguchi⁽¹⁾, and Kohei Hosokawa⁽¹⁾

⁽¹⁾ Connected Infrastructure Research Labs., NEC Corp., 1753 Shimonumabe, Nakahara-ku, Kanagawa, 211-8666 Japan, *shoma-tateno@nec.com

Abstract We propose a high-sensitivity, low-complexity FO-tolerant frame synchronisation scheme for low-SNR coherent optical inter-satellite links based on an amplitude-modulated training sequence and a multiplication-free amplitude-weighted timing metric, reducing the required overhead length by 37.5% and the computational complexity by 36.3% compared to conventional schemes. ©2026 The Author(s)

Introduction

The increasing demand for high-capacity long-distance optical inter-satellite links (OISLs) has driven the adoption of coherent systems because of their high sensitivity and spectral efficiency [1,2]. Severe geometric beam spreading causes significant path loss, requiring operation under low signal-to-noise ratio (SNR). In these conditions, pilot-aided digital signal processing (DSP) is preferred over blind approaches due to better sensitivity and stability [3-5]. Among the various receiver DSP functions, frame synchronisation (FS) is particularly critical, as any FS error directly undermines subsequent processing. Moreover, relative satellite motion induces Doppler shifts which necessitate a widely used two-stage frequency offset compensation (FOC) [2,6-8]. As a result, FS must operate reliably in the presence of a residual frequency offset (FO) after the initial FOC stage. Therefore, a robust and low-complexity FO-tolerant FS is essential for resource-limited coherent OISL systems [9].

To meet this requirement, several FO-tolerant FS schemes have been proposed [10-14]. However, they generally involve a trade-off between sensitivity and complexity. These methods can be broadly classified into two types. The first is an auto-correlation-based method; while offering relatively low computational complexity, it is susceptible to noise because correlations are computed between noisy received signals [10-12]. The second is a double cross-correlation-based method which improves noise robustness through successive cross-correlations, but at the cost of increased complexity [13,14]. While practically applicable to OISLs in their current form, further alleviating the trade-off between sensitivity and complexity remains highly beneficial for resource-limited OISL systems.

To overcome this trade-off, we propose a high-sensitivity and low-complexity FO-tolerant FS scheme that combines an auto-correlation framework with a specialised amplitude-modulated (AM) training sequence. Unlike conventional phase-shift-keying (PSK)-based approach, our

AM-based approach exploits both amplitude and phase profiles to reduce asynchronous similarity and suppress false FS detection even under low-SNR. To further enhance detection robustness, we incorporate a multiplication-free amplitude weighting into the timing metric (TM). Through simulations and experimental verification, we demonstrate that the proposed FO-tolerant FS method mitigates the conventional trade-off and achieves high sensitivity with low complexity. These results indicate that the proposed scheme provides a practical synchronisation solution for resource-limited coherent OISLs.

Principle and Model Design

Figure 1(a) describes our pilot-aided DSP flow in this paper. On the Tx DSP side, PRBS bits are mapped to the payload, overhead (OH) and pilot symbols. The OH symbols for FS and pre-equalization are inserted at the beginning of the frame, and pilot symbols are inserted into the payload with the period of L_p . These symbols are then upsampled and RRC-filtered (roll-off 0.5). We assume the transmission system where a laser control roughly compensates FO due to DS as the initial FOC. Thus, the received signal has some residual FO for the subsequent Rx DSP to work [6,7]. The received signal is RRC-filtered. After FS, the T-spaced pilot-aided 2×2 MIMO with PLL-based carrier phase recovery (CPR) performs

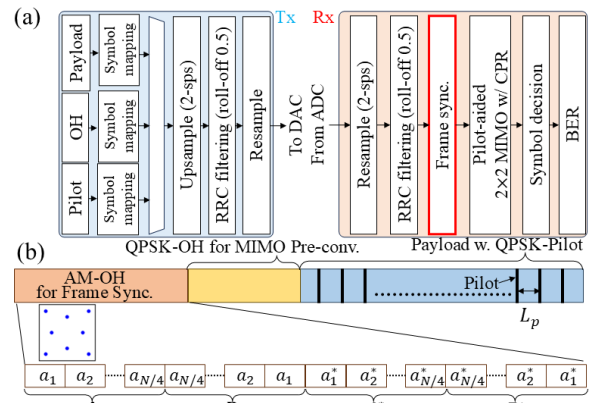


Fig. 1: (a) DSP flow. (b) Frame construction

Tab. 1: FO-tolerant FS algorithms. N is the number of OH symbols (referred to OH length)

Scheme	Ref.	OH sequence	OH format	TM type	Number of complex-multiplications per symbol
Case (1)	[10,11]	$[A, B, A^*, B^*]$	PSK	Auto-corr.	$N/2 + 1$
Case (2)	[12]	$[A, B^*, B^*, A]$	PSK	Dual-metric product auto-corr.	$N/2 + 2$
Case (3)	[13,14]	$[A]$	PSK	Double cross-corr.	N
Case (4)	This study	$[A, B, A^*, B^*]$	8QAM	Weighted auto-corr.	$N/2 + 1$

polarisation demultiplexing and compensates for the residual FO and laser phase noise [15]. Finally, the bit-error rate (BER) is evaluated.

The proposed FS method utilises an AM-OH sequence and an amplitude-weighted auto-correlation-based TM. The AM-OH has a centrally symmetric conjugate structure given by $[A, B, A^*, B^*]$, where $*$ denotes complex conjugate operation, B is the symmetric sequence of $A = [a(1), \dots, a(N/4)]$, and N is the OH length (Fig. 1(b)). Unlike conventional PSK OH, the AM-OH exploits both amplitude and phase profiles to reduce asynchronous similarity and false detection. The AM-OH symbols have the same mean power as the payload symbols. The OH-format optimisation is discussed in Fig. 2(a).

At the receiver, the frame starting point is identified as the peak position of the TM obtained by sliding an N -symbol window over one frame. In this study, we utilised the following TM, $M(d)$:

$$M(d) = \left| \sum_{k=0}^{N/2-1} R(d-k) \cdot R(d+k) \cdot s(k) \right|, \quad (1)$$

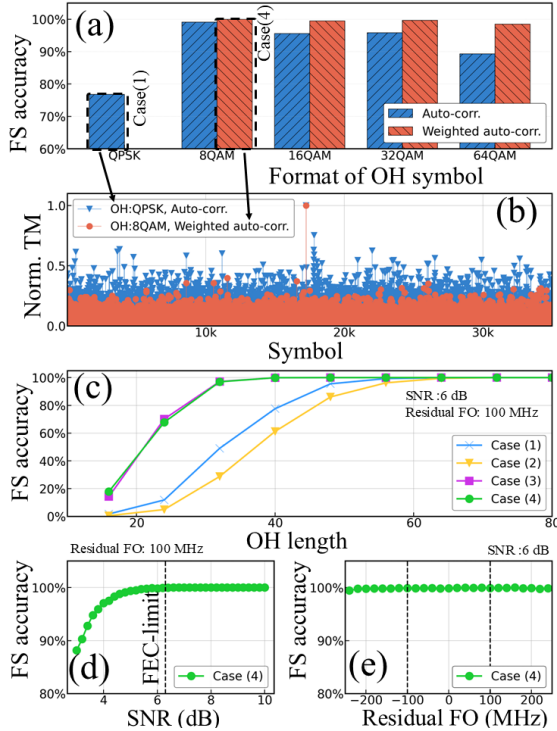


Fig. 2: (a) FS accuracy with different OH formats. (b) Norm. TM by peak of M . FS accuracy vs. (c) OH length, (d) SNR, and (e) residual FO.

where d is the time index of the 1-sps received symbol $R(d)$ of one polarization after RRC-filtering. The proposed method achieves FO robustness by utilising the centrally symmetric conjugate OH structure and a symmetric correlation. We introduce a weighting factor, $s(k)$, corresponding to the amplitude of the OH symbol, which de-emphasises noise-sensitive small-amplitude symbols and further improves FS sensitivity. Notably, for efficient implementation, we set $s(k)$ to a quantised value nearest to a power of two, which can be implemented using only bit shifts, yielding a multiplier-free weight operation.

Table 1 compares various FO-tolerant FS algorithms. Complexity is evaluated by the number of complex multiplications required for TM calculation without normalisation. Previous studies typically use PSK OH symbols, such as BPSK and QPSK. Case (1) is a standard auto-correlation-based FS algorithm exploiting a centrally symmetric conjugate OH structure [10,11]; case (2) utilises dual-metric product to suppress sidelobes [12]; and case (3) is a double cross-correlation scheme. Compared to cases (1,2), case (3) correlates a noise-free training symbol with the received signal, improving noise robustness at the cost of approximately twice the complexity.

Simulation results

We investigated the performance through numerical simulations using a PM-QPSK 1 Gbaud signal ($N_p = 16$). We set the SNR to 6 dB, near the forward error correction (FEC) limit, and the laser linewidth to 1 kHz. We applied a residual FO of 100 MHz (10% of the baud rate) in the received signal. This is near the upper limit of the operating range for the subsequent DSP flow [16] and is also reasonable after the preceding laser control stage [6,7]. The receiver SNR fading due to pointing jitter has a slower timing scale compared to DSP processing, so it can be assumed to be quasi-static [9]. This study therefore evaluated performance at a fixed SNR. Note that in actual operation, the SNR margin should be considered. We evaluated FS methods by analysing the FS accuracy over 10,000 simulation trials, where FS is judged correct when the detected frame starting point given by the peak point of M equals the true one. All FS schemes were evaluated under

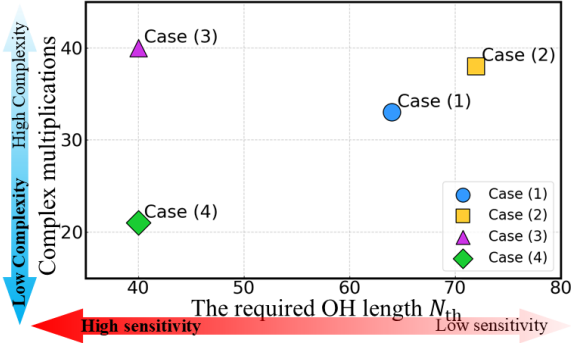


Fig. 3: The required OH length N_{th} and complex multiplications per symbol for calculating TM

the same signal and receiver conditions.

We first optimised the OH-format in Fig. 2(a), where N is set to 40. The AM format improves FS accuracy, and adding weighting coefficients, $s(k)$, further enhances it. This is clearly seen from TM (Fig. 2(b)), where peak-to-sidelobe ratios (PSRs) are 13.5 and 24.8, respectively. PSR, defined as $(M_{peak} - \mu)/\sigma$, indicates the quality of TM peak, where M_{peak} , μ and σ are the peak, mean and standard deviation of M . Among the tested AM formats, 8QAM was selected as the optimal format, offering the best trade-off between false-peak suppression and noise tolerance, showing the highest FS accuracy (Fig. 2(a)). Figure 2(c) shows FS accuracy vs. OH length. While the conventional approaches utilised QPSK for the FS OH symbols, the proposed method employed 8QAM. Thanks to the optimal 8QAM format and the amplitude-weighted TM, our proposed method demonstrated the highest FS accuracy despite relatively short OH lengths. We also evaluated the SNR and FO-tolerance (Figs. 2(d,e)). The proposed method exhibits robustness against residual FO within the operating range for the subsequent CPR, and provides a margin of 0.5 dB from the FEC limit when $N = 40$.

Figure 3 shows the required OH length N_{th} vs. the number of complex multiplications per symbol for calculating TM. N_{th} is given by the minimum OH length achieving an FS accuracy $\geq 99.9\%$ (Fig. 2(c)), and we count the number of complex multiplications from N_{th} and complexity formula (Tab. 1). Owing to its higher FS sensitivity, the proposed method reduces N_{th} , which in turn lowers the overall complexity. The proposed method achieves 37.5% reduction of N_{th} from auto-correlation-based approaches (cases (1-2)). The proposed algorithm also exhibits the lowest complexity, achieving a 36.3% reduction over conventional methods.

Experimental setup and results

Finally, to experimentally validate stable FS operation of the proposed method under low-SNR and residual-FO conditions relevant to coherent

OISLs, we performed a back-to-back experiment with the PM-QPSK 1 Gbaud signal (Fig. 4(a)). The frame structure is the same as in the simulations. Based on the simulation results, we utilised 40 symbols of 8QAM OH for FS. The same OH sequence was inserted on both polarisations to ensure that FS operates under polarisation mixing. On the Tx side, the electrical signal was modulated with a PM-IQ modulator and four 2 GSa/s digital-to-analog converters (DACs). The optical signal was attenuated by a variable optical attenuator (VOA), while monitoring the received optical power (ROP), and was captured by the coherent receiver and 2 GSa/s analog-to-digital converters (ADCs), and the electrical signals were processed offline as Rx DSP flow (Fig. 1). For both the signal source and local oscillator (LO), we utilised external cavity lasers (ECLs) of 1553.3 nm and 1-kHz linewidth. We tuned the ECL wavelength of LO to give a residual FO of approximately 100 MHz. Figure 4(b) shows BER vs. ROP obtained by adjusting the VOA. We present the constellation and TM at ROP of -53 dBm, i.e. near FEC-limit (Figs. 4(c,d)). PSR of Fig. 4(c) is 49.5, which is sufficiently high to ensure stable FS. These findings validates the effectiveness of the proposed method under receiver conditions relevant to low-SNR coherent OISLs.

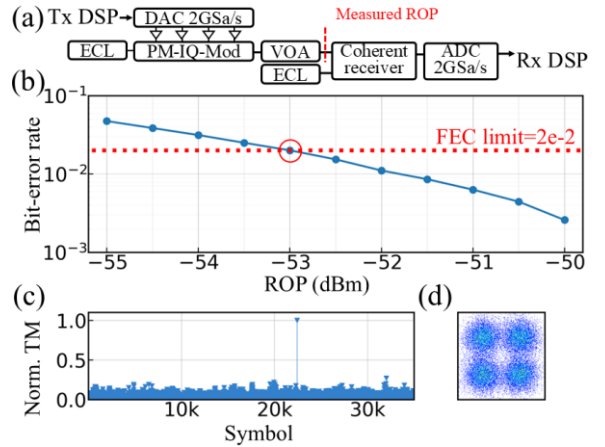


Fig. 4: (a) Experimental setup, (b) BER vs. ROP. (c) Norm. TM and (d) Constellation at ROP of -53 dBm.

Conclusions

We propose a high-sensitivity, low-complexity FO-tolerant FS scheme for coherent OISLs. By utilising an 8QAM OH sequence and a multiplication-free amplitude-weighted TM, our method mitigates the sensitivity–complexity trade-off, reducing N_{th} by 37.5% and the computational complexity by 36.3% compared to conventional approaches. Furthermore, experimental validation with a 1 Gbaud PM-QPSK signal confirmed stable operation near the FEC limit, achieving a high PSR of 49.5 at ROP of -53 dBm.

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